

INT 303 BIG DATA ANALYTICS

Lab 3: Exploratory Data Analysis (EDA)

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Lab: Exploratory Data Analysis (EDA)

Background

- The Data Science Process:

Ask an interesting question

Get the Data

Explore the Data

Model the Data

Communicate/Visualize the Results



Exploratory Data Analysis (EDA)

What?

- EDA encompasses the “*explore* data” part of the data science process
- EDA is crucial but often overlooked
- Explore **global** properties: use histograms, scatter plots, and aggregation functions to summarize the data
- Explore **group** properties: group like-items together to compare subsets of the data (are the comparison results reasonable/expected?)

EDA: without Pandas

top50.cs

- Each row represents a distinct song. The columns are:
- **ID**: a unique ID (i.e., 1-50)
- **TrackName**: Name of the Track
- **ArtistName**: Name of the Artist
- **Genre**: the genre of the track
- **BeatsPerMinute**: The tempo of the song.
- **Energy**: The energy of a song - the higher the value, the more energetic.
- **Danceability**: The higher the value, the easier it is to dance to this song.
- **Loudness**: The higher the value, the louder the song.
- **Liveness**: The higher the value, the more likely the song is a live recording.
- **Valence**: The higher the value, the more positive mood for the song.
- **Length**: The duration of the song (in seconds).
- **Acousticness**: The higher the value, the more acoustic the song is.
- **Speechiness**: The higher the value, the more spoken words the song contains.
- **Popularity**: The higher the value, the more popular the song is.

EDA: without Pandas

We have a small dataset of the top 50 most-streamed Spotify songs, globally, for 2019.

top50.csv

	TrackName	ArtistName	Genre	BeatsPer Minute	Energy	Danceability	Loudness	Live ness	Valence	Length	Acousticness	Speechi ness	Popularity
1	Senorita	Shawn Mendo	canadian pop	117	55	76	-6	8	75	191	4	3	79
2	China	Anuel AA	reggaeton flow	105	81	79	-4	8	61	302	8	9	92
3	boyfriend (w	Ariana Grande	dance pop	190	80	40	-4	16	70	186	12	46	85
4	Beautiful Pec	Ed Sheeran	pop	93	65	64	-8	8	55	198	12	19	86

⋮

- **Q1:** What are some ways we can store this file into data structure(s) using regular Python (not the Pandas library).

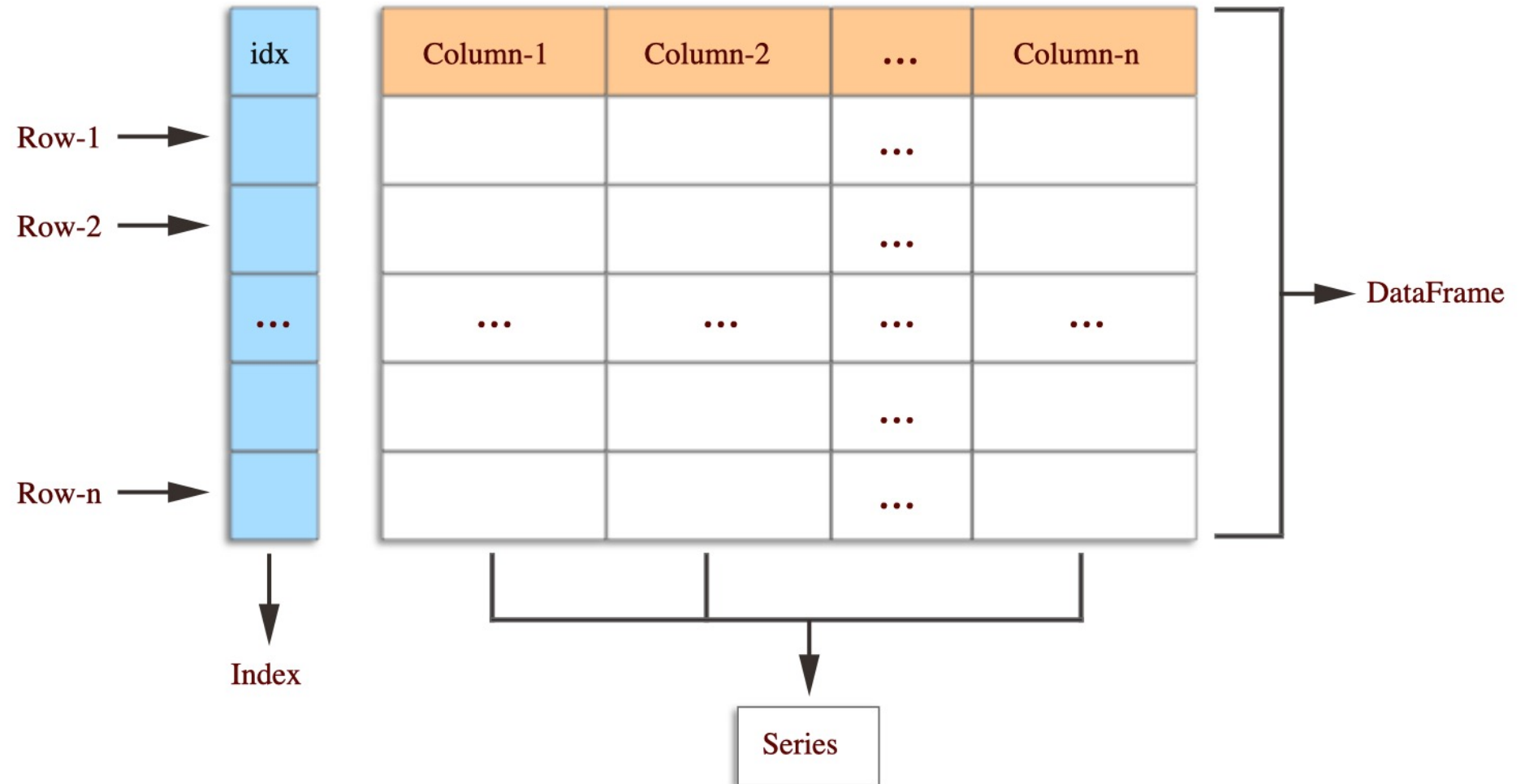
EDA: with Pandas!



Kung Fu Panda is property of DreamWorks and
Paramount Pictures

Pandas

Pandas Data structure



EDA: with Pandas

How

- import **pandas** library (convenient to rename it)
- Use **read_csv()** function

```
import pandas as pd  
dataframe = pd.read_csv("yourfile.csv")
```

EDA: with Pandas

Common Panda functions

- **High-level viewing:**
 - `head()` – first N observations
 - `tail()` – last N observations
 - `columns()` – names of the columns
 - `describe()` – statistics of the quantitative data
 - `dtypes()` – the data types of the columns

EDA: with Pandas

Common Panda functions

- **Accessing/processing:**

- `df["column_name"]`
- `Df.column_name`
- `.max()`, `.min()`, `.idxmax()`, `.idxmin()`
- `<dataframe> <conditional statement>`
- `.loc[]` – label-based accessing
- `.iloc[]` – index-based accessing
- `.sort_values()`
- `.isnull()`, `.notnull()`

EDA: with Pandas

Common Panda functions

- **Grouping/Splitting/Aggregating:**

- `groupby()`, `.get_groups()`
- `.merge()`
- `.concat()`
- `.aggregate()`
- `.append()`

Pandas example outlines

- Import Dataset
- Display Dataset
- Merge Dataset
- Rebuild Dataset
- Dataset transformation
- Group by
- Joint

Import Dataset

```
#importing the dataset by reading the csv file  
data = pd.read_csv(/content/Iris.csv)
```

```
#displaying the first five rows of dataset  
data.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

display First five rows of dataset

```
In [49]: dfcwci.head()
```

```
Out[49]:
```

	id	last_name	first_name	middle_name	street_1	street_2	city	state	zip	amount	date	candidate_id
0	NaN	Agee	Steven	NaN	549 Laurel Branch Road	NaN	Floyd	VA	24091	500.0	2007-06-30	16
1	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	250.0	2007-05-16	16
2	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	50.0	2007-06-18	16
3	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	100.0	2007-06-21	16
4	NaN	Akin	Charles	NaN	10187 Sugar Creek Road	NaN	Bentonville	AR	72712	100.0	2007-06-16	16

```
In [22]: del dfcwci['id']  
dfcwci.head()
```

```
Out[22]:
```

	last_name	first_name	middle_name	street_1	street_2	city	state	zip	amount	date	candidate_id
0	Agee	Steven	NaN	549 Laurel Branch Road	NaN	Floyd	VA	24091	500.0	2007-06-30	16
1	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	250.0	2007-05-16	16
2	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	50.0	2007-06-18	16
3	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	100.0	2007-06-21	16
4	Akin	Charles	NaN	10187 Sugar Creek Road	NaN	Bentonville	AR	72712	100.0	2007-06-16	16

How to get useful infos

```
# input
state = np.random.RandomState(100)
ser = pd.Series(state.normal(10, 5, 25))
```

```
# using pandas
ser.describe()
```

```
count    25.000000
mean     10.435437
std       4.253118
min       1.251173
25%       7.709865
50%      10.922593
75%      13.363604
max       18.094908
dtype: float64
```

Rebuild Dataset (1)

- To find and fill the missing data in the dataset we will use another function.
- **Using isnull() /isna() function:**

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False

Rebuild Dataset (2)

- Using `isna().sum()`

```
data.isna().sum()
```

```
Id          0  
SepalLengthCm  0  
SepalWidthCm  0  
PetalLengthCm  0  
PetalWidthCm  0  
Species      0  
dtype: int64
```

Rebuild Dataset(3)

- **DataFrame.fillna()**
- Fill NA/NaN values using the specified method.

pandas.DataFrame.fillna

`DataFrame.fillna(value=None, method=None, axis=None, inplace=False, limit=None, downcast=None)` [\[source\]](#)

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.fillna.html>

Rebuild Dataset (4)

- If a dataset contains duplicate values it can be removed using the `drop_duplicates()` function.

`pandas.DataFrame.drop_duplicates`

```
DataFrame.drop_duplicates(subset=None, keep='first', inplace=False,  
ignore_index=False)
```

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop_duplicates.html

GroupBy Dataset(1)

- Contributors

```
dfcwci=pd.read_csv("../data/contributors_with_candidate_id.txt", sep="|")  
dfcwci.head()
```

	id	last_name	first_name	middle_name	street_1	street_2	city	state	zip	amount	date	candidate_id
0	NaN	Agee	Steven	NaN	549 Laurel Branch Road	NaN	Floyd	VA	24091	500.0	2007-06-30	16
1	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	250.0	2007-05-16	16
2	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	50.0	2007-06-18	16
3	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	100.0	2007-06-21	16
4	NaN	Akin	Charles	NaN	10187 Sugar Creek Road	NaN	Bentonville	AR	72712	100.0	2007-06-16	16

GroupBy Dataset(2)

- Groupby:
 - Splitting the data into groups based on some criteria
 - Applying a function to each group independently
 - Combining the results into a data structure

```
In [28]: dfcwc.groupby("state").sum()
```

```
Out[28]:
```

	zip	amount	candidate_id
state			
AK	2985459621	1210.00	111
AR	864790	14200.00	192
AZ	860011121	120.00	37
CA	14736360720	-5013.73	600
CO	2405477834	-5823.00	111
CT	68901376	2300.00	35
DC	800341853	-1549.91	102
FL	8970626520	-4050.00	803

Merge Dataset

- Merging the dataset is the process of combining two datasets in one.

pandas.DataFrame.merge ¶

```
DataFrame.merge(right, how='inner', on=None, left_on=None, right_on=None,  
left_index=False, right_index=False, sort=False, suffixes=('_x', '_y'), copy=True,  
indicator=False, validate=None)
```

[\[source\]](#)

<https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.merge.html>

Merge Dataset

- Example: Candidates

```
In [8]: dfcand=pd.read_csv("../data/candidates.txt", sep='|')
dfcand.head(10)
```

```
Out[8]:
```

	id	first_name	last_name	middle_name	party
0	33	Joseph	Biden	NaN	D
1	36	Samuel	Brownback	NaN	R
2	34	Hillary	Clinton	R.	D
3	39	Christopher	Dodd	J.	D
4	26	John	Edwards	NaN	D
5	22	Rudolph	Giuliani	NaN	R
6	24	Mike	Gravel	NaN	D
7	16	Mike	Huckabee	NaN	R
8	30	Duncan	Hunter	NaN	R
9	31	Dennis	Kucinich	NaN	D

```
dfcwci=pd.read_csv("../data/contributors_with_candidate_id.txt", sep="|")
dfcwci.head()
```

	id	last_name	first_name	middle_name	street_1	street_2	city	state	zip	amount	date	candidate_id
0	NaN	Agee	Steven	NaN	549 Laurel Branch Road	NaN	Floyd	VA	24091	500.0	2007-06-30	16
1	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	250.0	2007-05-16	16
2	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	50.0	2007-06-18	16
3	NaN	Ahrens	Don	NaN	4034 Rennellwood Way	NaN	Pleasanton	CA	94566	100.0	2007-06-21	16
4	NaN	Akin	Charles	NaN	10187 Sugar Creek Road	NaN	Bentonville	AR	72712	100.0	2007-06-16	16

Merge Dataset(2)

- Merge:
- Combine tables on a common key-value

```
In [40]: cols_wanted=['last_name_x', 'first_name_x', 'candidate_id', 'id', 'last_name_y']  
dfcwc.merge(dfcand, left_on="candidate_id", right_on="id")[cols_wanted]
```

```
Out[40]:
```

	last_name_x	first_name_x	candidate_id	id	last_name_y
0	Agee	Steven	16	16	Huckabee
1	Akin	Charles	16	16	Huckabee
2	Akin	Mike	16	16	Huckabee
3	Akin	Rebecca	16	16	Huckabee
4	Aldridge	Brittni	16	16	Huckabee

Question ---Mapping

Following is a preview of the DataFrame `df` :

x	y	z
1	NaN	1
2	NaN	2
NaN	1	3

Match the commands to their expected outputs:

1 `df.notna().sum()`

2 `df.isna().any()`

3 `df['z'].isna()`

```
# A
0    False
1    False
2    False
Name: z, dtype: bool
```

```
# B
x      2
y      1
z      3
dtype: int64
```

```
# C
x      True
y      True
z     False
dtype: bool
```

Grammar of Data

- If you need to find a Data Related work!!!
- <https://www.w3resource.com/python-exercises/pandas/index.php>



THANK YOU



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